DIT349: Fundamentals of Software Architecture HT25 Final Exam

Time: 14:00-18:00
Examiner: Krzysztof Wnuk 0 706 122 044
Place: October 27, 2025, 14.00-18.00 at Lindholmen
I'm available for questions at 15.00 and 16.30 during the exam time
Max Score: 100
Exam aids: none (except for generally allowed aids, such as dictionaries)

Grading Scale: 3: >=50 4: >=70 5: >=85
The exam consists of the following parts:

Table of Contents

Sheet to clarify the notation for component diagrams in UML and C4 notation:..... 2

P1: Architectural drivers, quality attributes and architectural significance (20p) ... 5

P2: Architectural Tactics (16p)..... 6

P3: Architectural styles and architecture modeling (22p) 7

P4: Software Architecture Principles (18p)..... 8

P5: Fill the Gap (24p)..... 12

Answer in full sentences or paragraphs in questions where a description, explanation or similar is required. Please write legibly. If we cannot read your handwriting, we cannot give you points.

Read each assignment thoroughly before starting to work on it. Begin each assignment on a new sheet. Only write on the front of each sheet.
Label each sheet with:

- The assignment number and sub-assignment number (e.g., P1.A, …)
- The anonymous code provided by the student office. (The exam is anonymous.)

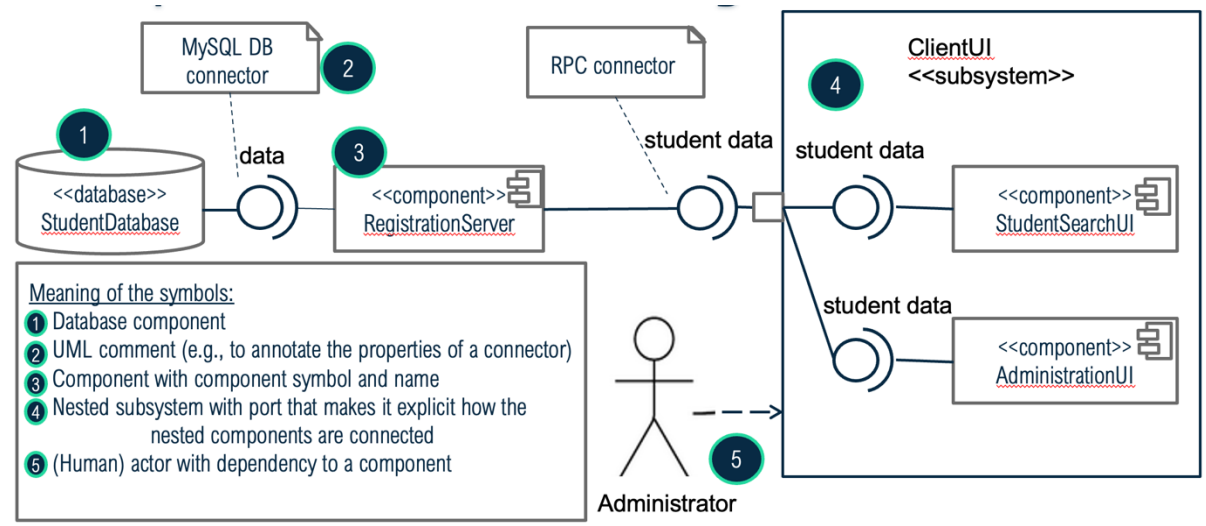
Before handing it in: Sort your sheets in the assignment order and enumerate them as 1, 2, 3, …

Additional information

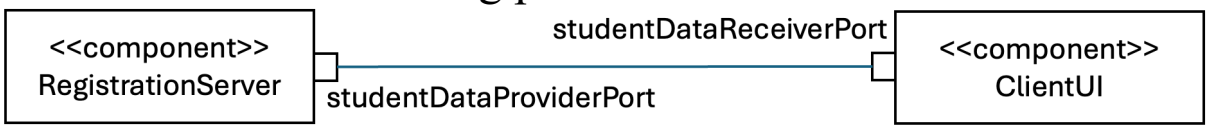
Keep in mind that we always require you to motivate your answer and to demonstrate your understanding of the subject matter.

Good luck!

Sheet to clarify the notation for component diagrams in UML and C4 notation:

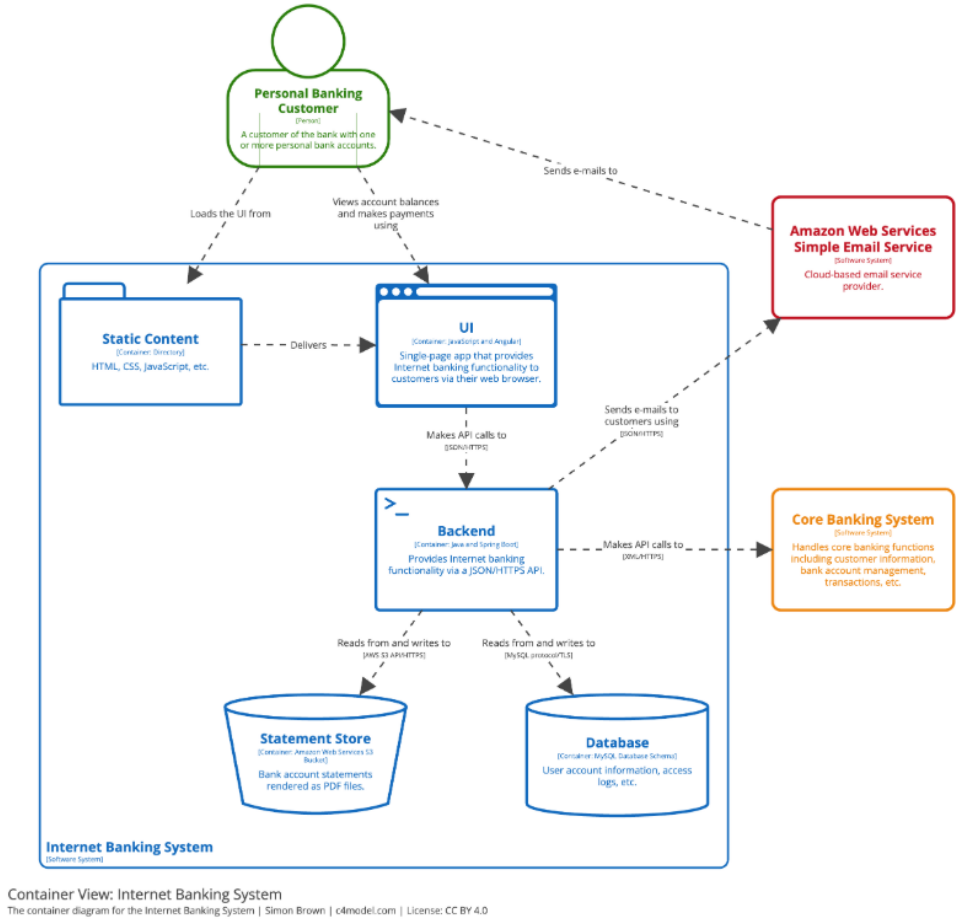


Alternative notation using ports:



Sheet to clarify the notation for C4 diagrams

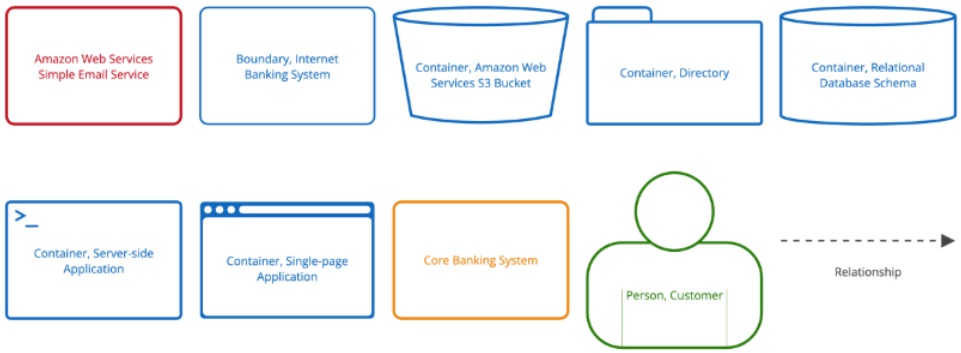
Example container diagram



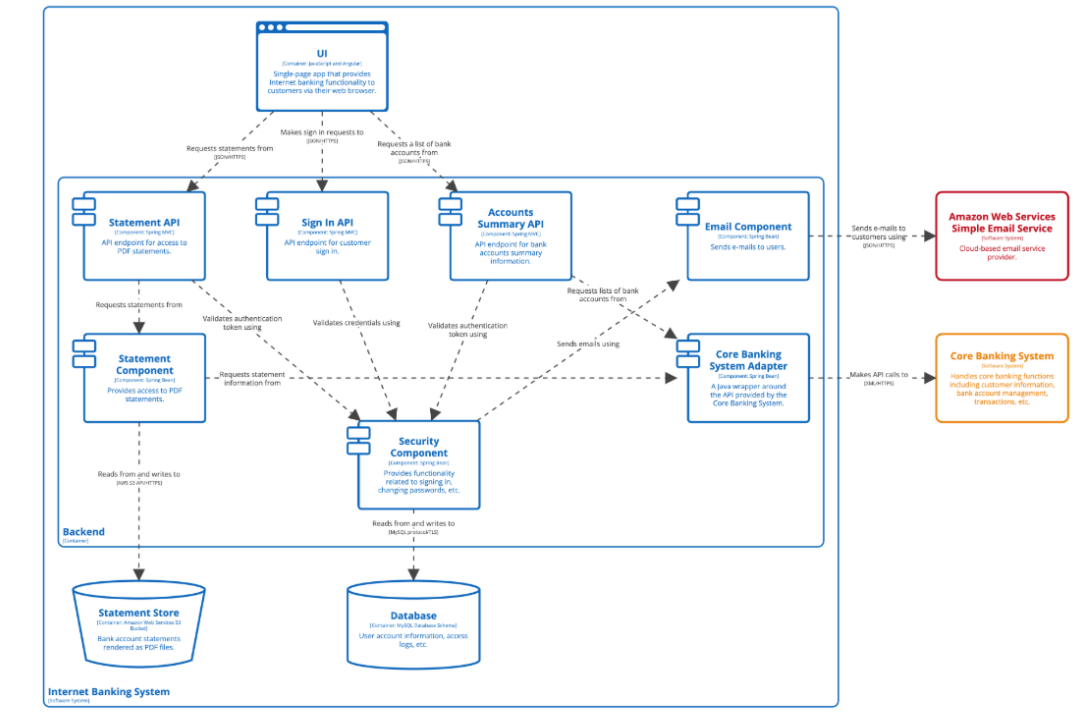
Container View: Internet Banking System

The container diagram for the Internet Banking System | Simon Brown | c4model.com | License: CC BY 4.0

Diagram key

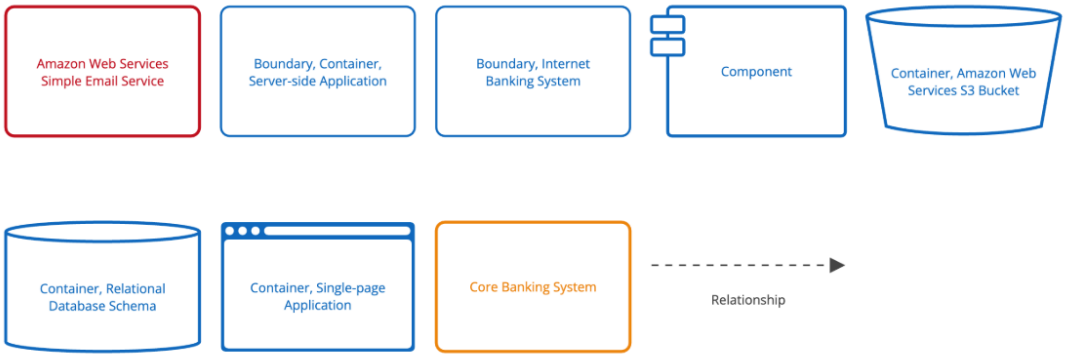


Example component diagram



Component View: Internet Banking System - Backend

The component diagram for the Internet Banking System Backend | Simon Brown | c4model.com | License: CC BY 4.0



P1: Architectural drivers, quality attributes and architectural significance (20p)

A. List three architectural drivers that we learned in the course and explain them briefly. (3p)
B. We learned that constraints belong to two categories. What are those two categories of constraints that influence the architecture? Give one example constraint for each category. For each of the examples, explain why it is an architectural driver. (6p)
C. Briefly define what “architecturally significant requirements” are. Give one example of a requirement that is not architecturally significant. (3p)
D. What is a quality attribute scenario? Provide an example quality attributes scenario (with all six parts) (5p)
E. Provide an example of an architecture trade-off (3 points)

P2: Architectural Tactics (16p)

A. Provide the definition of an architecture tactic, discuss the relation between the architecture tactic and architecture style (2p)

B. Fill in the related quality attribute next to the tactic (tip, each quality attribute has two tactics in the table below) (10p)

Tactic	Quality attribute (select between interoperability, security, modifiability, availability, testability)
Split module	
Encapsulate	
Heartbeat	
Active redundancy	
Discover service	
Orchestrate	
Detect intrusion	
Authenticate actors	
Sandbox	
Limit non-determinism	

C. What is the architecture style/pattern that is suitable for monolithic deployment and why? (2p)

D. What is the architecture style/pattern that is suitable for small, simple applications and websites and why? (2p)

P3: Architectural styles and architecture modeling (22p)

You want to design designing a secure and reliable online auction platform for used cars, accessible via both a website and a mobile app. The platform shall support a range of functionalities tailored to buyers and sellers, including the ability for sellers to list vehicles and for buyers to place bids in real time. Buyers will be able to communicate with the auction house through a live chat to ask questions about specific vehicles, while a marketplace interface will allow users to browse ongoing and upcoming auctions. A secure payment system should enable buyers to deposit funds and sellers to withdraw earnings safely. Sellers can enhance their listings by uploading car service history documents, technical overview videos, condition summaries, and certified mechanic reports detailing potential repairs or issues. Buyers should benefit from comparison tools that allow them to evaluate listed vehicles against similar models available online to better understand market value and pricing. Additionally, the platform shall support trade-in functionality, enabling sellers to exchange their current vehicle when purchasing another through the system.

Your task:

A. Describe what architectural style or combination of styles you think would be most appropriate for this system. Briefly motivate your choice. (4p)
B. Draw one UML component and connector diagram or draw two C4 diagrams (one container diagram and one component diagram) that show how you plan to design the architecture of the system. You should do either C4 or UML notation, not both! (15p)
C. Describe what three quality attributes you think are the most important. List one architecture tactic for each of the selected quality attributes (3p)

P4: Software Architecture Principles (18p)

Below you see 2 examples of Java code that clearly violate one of the architecture principles that we discussed in this course (each example violates one principle). Please analyze this code and provide answers to the following questions.

Example A is a banking application where you can do transactions on your bank account.
Example B is – a user registration system for an application. The main functionality is that the system prompts the user to enter a username, password and email through console input.

Please provide answers to the following questions:

A. What is the architecture principle that the example system A is violating? (1p)
B. Comment in 5 places in the example Java code for system A where this principle is violated and explain why it is violated (5p). The code has more than 5 issues that violate this architecture principle.
C. Suggest ways to improve the code to follow the software architecture principle. You do not need to write the correct code, just suggest the improvement and point to the place where it should be implemented (you can comment directly in the example code below or write a summary paragraph below the code, use comments or pseudo-code if you want, please refer to the lines on the left side of the code) (3p)
D. What is the architecture principle that the example system B is violating? (1p)
E. Comment in 5 places in the example Java code for system B where this principle is violated and explain why it is violated (5p). The code has more than 5 issues that violate this architecture principle.
F. Suggest ways to improve the code in example B to follow the software architecture principle. You do not need to write the correct code, just suggest the improvement and point to the place where it should be implemented (you can comment directly in the example code below or write a summary paragraph below the code) (3p)

Example Java Code for System A

```
1 class BankAccount {
2   public String accountNumber;
3   public double balance;
4   public String[] transactionHistory;
5   public int transactionCount;
6   public double dailyWithdrawalLimit;
7   public double todayWithdrawnAmount;
8   // Constructor
9   public BankAccount(String accountNumber, double initialBalance) {
10      this.accountNumber = accountNumber;
11      this.balance = initialBalance;
12      this.transactionHistory = new String[100];
13      this.transactionCount = 0;
14      this.dailyWithdrawalLimit = 1000.0;
15      this.todayWithdrawnAmount = 0.0;
16  }
17 }
```

```
// Client code
class BankingApp {
  public static void main(String[] args) {
      BankAccount account = new BankAccount("123456", 5000.0);
      account.balance = 1000000.0; // Instant millionaire!
      account.todayWithdrawnAmount = 0.0; // Reset daily limit
      account.transactionCount = -5; // Invalid state
      account.transactionHistory[0] = "Hacked transaction";
      account.dailyWithdrawalLimit = Double.MAX_VALUE;
      System.out.println("Balance: " + account.balance);
  }
}
```

Example System B written in Java that manages the user registration of an application

```
import java.sql.*;
import java.util.Scanner;
class UserManager {

  private String dbUrl = "jdbc:mysql://localhost:3306/mydb";
  private String dbUser = "root";
  private String dbPassword = "password";

  public void handleUserRegistration() {
      Scanner scanner = new Scanner(System.in);
      System.out.println("=== User Registration ===");
      System.out.print("Enter username: ");
      String username = scanner.nextLine();
      System.out.print("Enter password: ");
      String password = scanner.nextLine();
      System.out.print("Enter email: ");
      String email = scanner.nextLine();

    if (username.length() < 3) {
          System.out.println("ERROR: Username too short!");
        return;
    }
    if (!email.contains("@")) {
        System.out.println("ERROR: Invalid email format!");
        return;
    }

    Connection conn = null;
    PreparedStatement stmt = null;
    try {
        Class.forName("com.mysql.jdbc.Driver");
        conn = DriverManager.getConnection(dbUrl, dbUser, dbPassword);

        String sql = "INSERT INTO users (username, password, email, created_at) VALUES (?, ?, ?,
NOW())";
        stmt = conn.prepareStatement(sql);
```

```
31          stmt.setString(1, username);
32          stmt.setString(2, password);
33          stmt.setString(3, email);
34            int rows = stmt.executeUpdate();

35          if (rows > 0) {
36                System.out.println("SUCCESS: User registered!");
37                System.out.println("Sending welcome email to: " + email);
38                sendEmail(email, "Welcome!", "Thanks for registering!")
39                System.out.println("[LOG] New user registered: " + username + " at " + new java.util.Date());
40                updateUserStatistics();
41               System.out.println("Total users: " + getTotalUsers());
42          }

43      } catch (SQLException e) {
44          System.out.println("DATABASE ERROR: " + e.getMessage());
45          e.printStackTrace();
46      } catch (ClassNotFoundException e) {
47          System.out.println("DRIVER ERROR: " + e.getMessage());
48      } finally {
49          try {
50              if (stmt != null) stmt.close();
51              if (conn != null) conn.close();
52          } catch (SQLException e) {
53              System.out.println("ERROR closing connection: " + e.getMessage());
54          }
55      }
56  }

57  private void sendEmail(String to, String subject, String body) {
58        System.out.println("Connecting to SMTP server...");
59       System.out.println("Sending email...");
60  }

61  private void updateUserStatistics() {
62          System.out.println("Updating statistics dashboard...");
63  }

64  private int getTotalUsers() {
65      try {
66              Connection conn = DriverManager.getConnection(dbUrl, dbUser, dbPassword);
67              Statement stmt = conn.createStatement();
68             ResultSet rs = stmt.executeQuery("SELECT COUNT(*) FROM users");
69          if (rs.next()) {
70              return rs.getInt(1);
71          }
72      } catch (SQLException e) {
73          System.out.println("Error getting user count");
74      }
75      return 0;
76  }
77   }
```

```
78  public class Main {
79     public static void main(String[] args) {
80            UserManager manager = new UserManager();
81           manager.handleUserRegistration();
82     }
83 }
```

P5: Fill the Gap (24p)

Write the letters (A, …, Q) on a sheet of paper and then list the answer(s) to the gaps.

A. The software architecture of a computing system is the set of _____ needed to reason about the system, which comprise _____, _____ among them and properties of both. (3p)
B. Modules represent a _____ way of considering the system. (1p)
C. Architecture focuses on interactions among the parts while programming focuses on _____ of parts. (1p)
D. Architectural drivers include _____ requirements, _____ requirements, and _____ (3p)
E. Constraints describe context that is _____ the architect's control (1p)
F. A tactic is a _____ decision that refines an architecture carried out in a style and is influential in controlling a quality attribute response (1p).
G. _____ is concerned with system failure and its associated consequences. A system failure occurs when a system no longer delivers a service that is consistent with its specification. (1p)
H. _____ is about the degree to which two or more systems can usefully exchange meaningful information via interfaces in a particular context (1p).
I. _____ is about protecting data and information from unauthorized access while still providing access to people and systems that are authorized (1p)
J. _____ is concerned with how easy it is for the user to accomplish a desired task and the kind of user support the system provides (1p)
K. An architecture quantum includes all the necessary _____ to function independently from other parts of the architecture (1p)
L. A _____ defines a point of potential interaction of a component with its environment (1p)
M. An architect must document software intensive systems from at least three perspectives, _____, _____ and _____ (3p)
N. _____ view shows how components are deployed to different devices/other physical entities (1p)
O. An _____ is a boundary across which two independent entities interact or communicate (1p)
P. There are two types of architecture reviews, _____ review, and _____ review (2p)
Q. The software architecture principle of _____ says that all information about a module should be private to the module unless required externally (1p)